



Theoretical study of T shaped phenothiazine/carbazole based organic dyes with naphthalimide as π -spacer for DSSCs

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ABSTRACT

Eight novel T shaped phenothiazine/carbazole based organic dyes with naphthalimide as π -spacer were designed, and the geometries, electronic structures, and optical features of these isolated dyes and dye-(TiO₂)₉ systems were investigated with density functional theory (DFT) and time dependent density functional theory (TD-DFT) calculations. Some quantify factors influencing the energy conversion efficiency (PCE) such as the light harvesting efficiency (LHE), electron injection driving force (ΔG_{inject}) and dye regeneration driving force (ΔG_{reg}) were also calculated for dye-sensitized solar cells (DSSCs) applications. It is found that these dyes show a good performance of electron injection and dye regeneration owing to the proper value of ΔG_{inject} and ΔG_{reg} . The optimized geometries of the non-planar molecular configuration of donor and the planar structure of the naphthalimide conjugated bridge are beneficial to efficient intramolecular charge transfer and the suppression of molecular aggregation. The properties about the electronic structure and absorption spectra indicate that replacement of benzene with thiophene unit near to cyanoacetic acid acceptor can generate more efficient conjugation effect and achieve red shift of absorption spectra, resulting a higher J_{sc} and V_{oc} in DSSCs device. The theoretical results reveal that DTPH2, DTPH4, DTCA2 and DTCA4 would be used as potential sensitizers for DSSCs applications.

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1. Introduction

Energy is the source power of social development. With the rapid development of economy and society, natural fossil fuels are exhausted day by day, and energy problem has become one of the universal concerned issues. Solar energy is an inexhaustible green energy with advantages of non-polluting, reliable and abundant supply, so solar energy is one of the best candidates for solving the energy crisis. In 1954, Bell Laboratory invented the silicon solar cells and pioneered in the development of solar energy by photo-electric conversion methods [1]. Since then, photovoltaic solar cells have been developed extensively to meet the world energy demand. In recent years, dye-sensitized solar cells (DSSCs) have become one of the most active fields in photovoltaic research owing to their low cost, facile fabrication, and high power conversion efficiencies (PCE).

In DSSCs, the dye sensitizers absorb sunlight, and inject photo-electrons to the conduction band of titanium dioxide (TiO₂) semiconductor, so the dye sensitizers play a crucial role in high PCE. In recent years, metal organic complexes and metal-free organic dye sensitizers have been widely developed and studied in DSSCs field, and metal-

free organic dyes have been considered to be promising candidates for DSSCs due to their simple synthesized process, higher molar extinction coefficient, easily tunable optical properties and environmental friendliness [2]. Among these metal free sensitizers, phenothiazine based organic dyes have been extensively studied and achieved impressive photovoltaic performance [3–10]. The non-planar butterfly conformation of the phenothiazine ring can sufficiently suppress molecular aggregation and the formation of molecular excimers, which is favorable for achieving high photo-voltage. On the other hand, the electron-rich sulfur and nitrogen heteroatom render phenothiazine a very strong electron donating ability [4,7,10]. In very recent, we reported the design and theoretical study of five phenothiazine-based organic molecules (DTPH1–5) with T shaped conformations, in which two 2-(hexylthio) thiophene units used as auxiliary electron donors were linked to C (3) and C (7) positions of phenothiazine ring, and different π -conjugated spacers were directly linked to N (10) position of the phenothiazine ring [11]. The optimized geometries indicate that these T shaped dyes show non-planar conformations, which are helpful in suppressing the close intermolecular π - π aggregation in device and enhancing thermal stability. Furthermore, the theoretical results reveal that these dyes could be used as potential sensitizers for DSSCs.

In order to enrich the research field of organic dyes for DSSCs and gain more information on the structure–property relationships, as a

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